

05_List_Comprehensions

July 3, 2023

List comprehensions are a way to write at the same way a loop and conditions and store results on a new list

```
[2]: mylist=['ford', 'bmw', 'ferrari', 'tesla', 'lamborghini']

new_list=[]
for car in mylist:
    initial=car[0]
    new_list.append(initial)

print(new_list)
```

```
['f', 'b', 'f', 't']
```

I can rewrite this as follows

```
[3]: new_list= [car[0] for car in mylist ]

print(new_list)
```

```
['f', 'b', 'f', 't']
```

```
[4]: #other example. put all in capital letters

new_list= [car.upper() for car in mylist ]

print(new_list)
```

```
['FORD', 'BMW', 'FERRARI', 'TESLA']
```

I can add a condition to put it in my list. fo instance, only add cars with more than 4 letters

```
[5]: new_list= [car for car in mylist if len(car)>4]

print(new_list)
```

```
['ferrari', 'tesla']
```

and finally, i can do different things to the elements depending on an if else condition. Inthis case, the if else condition apperas BEFORE the loop. For instance, if the car starts by F, we write in capital letter, and if it does not, we write in lowercase

```
[14]: new_list= [ car.upper() if car[0]=='f' else car for car in mylist]
```

Another example. you have a list with all the numbers from 1 to 10. 1- Make a new list in wwith only th eeven numbers 2- make a list i which all odd numbers are set to zero

```
[15]: list0=list(range(1,11))  
list0
```

```
[15]: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

```
[16]: list1=[x for x in list0 if x%2==0]  
list1
```

```
[16]: [2, 4, 6, 8, 10]
```

```
[17]: list2=[0 if x%2==0 else x for x in list0]  
  
print(list2)
```

```
[1, 0, 3, 0, 5, 0, 7, 0, 9, 0]
```

```
[ ]:
```